Introduction to Smooth Manifolds, Lee*
Solutions

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Problem 4-4. Let $\gamma(t) = (e^{2\pi i t}, e^{2\pi i \alpha t})$ for $\alpha \in \mathbb{R}\backslash\mathbb{Q}$. It suffices to show that $\{e^{2\pi i \alpha n} | n \in \mathbb{N}\}$ is dense in $\mathbb{S}^1$ since $\gamma(t+n) = \gamma(t) \cdot (1, e^{2\pi i \alpha n})$. This follows from the classical fact of analysis that the fractional parts of αn are dense in $[0,1]$ as n ranges over non-negative integers. For completeness, we indicate the proof.

Divide $[0,1]$ into N equal length intervals for N large and consider $0, \alpha, 2\alpha, \ldots, N\alpha$, which are distinct mod 1. Thus two, say $n\alpha$ and $m\alpha$ lie in the same subinterval. Then $|(n-m)\alpha \mod 1| < 1/N$, so every length $1/N$ subinterval contains some multiple of $(n-m)\alpha$, proving that $\{\alpha n \mod 1\}_{n \geq 0}$ is dense in $[0,1]$.

Problem 5-5. From Problem 4-4, $S = \gamma(\mathbb{R}) \subset \mathbb{T}^2$ is dense. S is obviously not an embedded submanifold because we cannot find a slice chart $\varphi : U \subset \mathbb{R}^2$ such that $\varphi(S)$ so that $\varphi(S)$ lies on the x-axis, since φ would be a homeomorphism onto its image and hence preserve denseness of $S \cap U$ in U.

Problem 5-6. If $M \subset \mathbb{R}^n$ is an embedded m-dimensional submanifold, define the unit tangent bundle $UM \subset TM$ as the set of all unit tangent vectors to M:

$$UM = \{(x,v) \in T\mathbb{R}^n | \, x \in M, v \in T_x M, |v| = 1\}.$$

UM is an embedded $(2m-1)$-dimensional submanifold of $T\mathbb{R}^n \approx \mathbb{R}^n \times \mathbb{R}^n$ since it is the level set of the norm function $|\cdot| : TM \to \mathbb{R}$ acting on the tangent space coordinate $|(x,v)| = |v|$. 1 is clearly a regular value, so UM is a codimension 1 properly embedded submanifold of TM by the regular level set theorem.

Problem 6-2. Whitney Immersion Theorem: Every smooth n-manifold with empty boundary admits a smooth immersion into $\mathbb{R}^{2n}$.

By Whitney embedding we can assume M is an embedded n-dimensional submanifold of $\mathbb{R}^{2n+1}$. Let $UM \subset T\mathbb{R}^{2n+1}$ be the unit tangent bundle of M and let $G : UM \to \mathbb{RP}^{2n}$ be projection $(x,v) \mapsto [v]$. UM is $(2n-1)$-dimensional so by Sard's theorem $G(UM)$ has

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†Solutions for Problem 8-2, 8-10, and 8-29 are by Sriram Bharadwaj.

measure 0 in $\mathbb{RP}^{2n}$. Therefore there exists $v \in \mathbb{R}^{2n+1} \setminus \mathbb{R}^{2n}$ that is not in the image of G. The projection $\pi : \mathbb{R}^{2n+1} \to \mathbb{R}^{2n}$ with kernel $\mathbb{R}v$ restricts to a smooth map of M into $\mathbb{R}^{2n}$. This is an immersion since $v \notin T_pM$ for any $p \in M$.

Problem 7-11. The Hopf action $\theta : \mathbb{S}^1 \times \mathbb{S}^{2n+1} \to \mathbb{S}^{2n+1}$ is smooth since it is just scalar multiplication on $\mathbb{C}^{n+1}$. If $w \in \mathbb{S}^{2n+1}$ then the orbits $\mathbb{S}^1 \cdot w$ are disjoint unit circles by definition of the action and the fact that orbits are disjoint or equal. Hence $w \in \mathbb{S}^{2n+1}$.

$$\coprod_w \mathbb{S}^1 \cdot w = \mathbb{S}^{2n+1}$$

Remark: For $n = 1$ the projection onto the orbit space $\pi : \mathbb{S}^3 \to \mathbb{S}^2$ is the Hopf fibration, with $\mathbb{S}^1$ fibres. We can construct an explicit map

$$\pi(w_1, w_2) = (2w_1\overline{w_2}, |w_1|^2 - |w_2|^2).$$

Problem 7-17. Considering the matrix groups as subsets of $\mathbb{R}^{n^2}$ or $\mathbb{C}^{n^2}$ we see that $GL(n, \mathbb{R})$ and $GL(n, \mathbb{C})$ are open as they are the preimages of $\mathbb{R} - \{0\}$ and $\mathbb{C} - \{0\}$ resp. under the determinant map (in particular they are not compact). $SL(n, \mathbb{R})$ and $SL(n, \mathbb{C})$ are closed but not bounded. For example, take the identity matrix with two of its diagonal entries replaced with x and $1/x$ for large $x \in \mathbb{R}$. $U(n) = \det^{-1}(\mathbb{S}^1)$ is compact since $\mathbb{S}^1$ is. $SU(n)$ is a closed subgroup of $U(n)$ and hence compact.

Problem 7-19. Suppose $G \cong N \rtimes_\theta H$ where $\theta : H \times N \to N$ is an action by automorphisms on N. Let $\varphi : G \to H$ be projection onto the second coordinate and let $\psi : H \to G$ be inclusion $h \mapsto (e, h)$. Then we have $\varphi \circ \psi = Id_H$ and $\ker \varphi = N$. Conversely, suppose such homomorphisms φ, ψ are given. Let $\ker \varphi = N$ and $H' = \psi(H)$. Suppose $h' \in \psi(h)$ and $h' \in \ker \varphi$. Then $h' = \varphi(\psi(h')) = \varphi(h') = e$. Hence $N \cap H' = \{e\}$. Moreover, $(\varphi \circ \psi \circ \varphi)(g) = \varphi(g)$ for any $g \in G$. We have $g = n\psi(\varphi(g))$ for some $n \in N$ so $NH' = G$. Therefore Theorem 7.35 shows $G = N \rtimes H'$.

Problem 8-2. We can differentiate the equation $\mathbf{f}(\lambda\mathbf{x}) = \lambda^c\mathbf{f}(\mathbf{x})$ to obtain:

$$D\mathbf{f}(\lambda\mathbf{x}).\mathbf{x} = c\lambda^c\mathbf{f}(\mathbf{x})$$

. Now, we make the following change of variables $x \mapsto x/\lambda$ to obtain:

$$D\mathbf{f}(\mathbf{x}).\mathbf{x} = c\lambda^c\mathbf{f}\left(\frac{\mathbf{x}}{\lambda}\right) = cf(\mathbf{x}).$$

Now, note that

$$\frac{\partial}{\partial x^j} f^i(\mathbf{x})\mathbf{e}_i = \frac{\partial f^i(\mathbf{x})}{\partial x^j}\mathbf{e}_i.\frac{\partial \mathbf{x}}{\partial x^j} = \frac{\partial f^i}{\partial x^j}x^j\mathbf{e}_i.\mathbf{e}_j = 0,$$

where $i \neq j$, $\mathbf{f} = (f^1, f^2, ..., f^n)$, $\mathbf{x} = (x^1, x^2, ..., x^n)$ and $\{\mathbf{e}_i\}_{i=1}^n$ is the standard basis for $\mathbb{R}^n$.

Problem 8-10. Because of the domain of definition of F, its inverse exists and is given by:

$$F^{-1}(u, v) = \left(\sqrt{\frac{u}{v}}, \sqrt{uv}\right).$$

This map is clearly differentiable in its domain of definition. The push-forward is:

$$F_*X(u,v) = dF(F^{-1}(u,v)).X(F^{-1}(u,v)) = \begin{pmatrix} \sqrt{uv} & \sqrt{\frac{u}{v}} \\ -\sqrt{\frac{v^3}{u}} & \sqrt{\frac{v}{u}} \end{pmatrix} \begin{pmatrix} \sqrt{\frac{u}{v}} \\ \sqrt{uv} \end{pmatrix} = \begin{pmatrix} 2u \\ 0 \end{pmatrix}.$$

Computed in exactly the same way, we have:

$$F_*Y(u,v) = \begin{pmatrix} uv \\ -v^2 \end{pmatrix}.$$

Problem 8-13. Let $X = \partial_x$ be the coordinate vector field on $\mathbb{R}^2$. We pullback by stereographic projection $\sigma : \mathbb{S}^2 \setminus \{N\} \to \mathbb{R}^2$ where N is the north pole. Let $Y = (\sigma^{-1})_*X$. and define $Y(N) = 0$. Y is clearly nonvanishing on $\mathbb{S}^2 \setminus N$. To check it is smooth we compute the coordinate representation of Y in the chart $(\tau, \mathbb{S} \setminus \{S\})$ where τ is projection from the south pole $S = (0,0,-1)$. Note that

$$\tau \circ \sigma^{-1}(x,y) = \sigma \circ \tau^{-1}(x,y) = \frac{(x,y)}{x^2+y^2}$$

is the circle inversion about the unit circle in $\mathbb{R}^2$. A calculation shows

$$D(\tau \circ \sigma^{-1})(\sigma \circ \tau^{-1}(x,y)) = \begin{pmatrix} x^2-y^2 & -2xy \\ -2xy & y^2-x^2 \end{pmatrix}$$

so in this chart Y is given by

$$D(\tau \circ \sigma^{-1})_{\sigma\tau(x,y)} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} x^2-y^2 \\ y^2-x^2 \end{pmatrix}$$

which vanishes precisely at $\tau(N) = (0,0)$.

Problem 8-23. The evaluation maps as in Proposition 8.27 give isomorphisms $\varepsilon_1 : \mathrm{Lie}(G) \to T_eG$, $\varepsilon_2 : \mathrm{Lie}(H) \to T_eH$ and $\varepsilon : \mathrm{Lie}(G \times H) \to T_{(e,e)}(G \times H)$. We have the vector space isomorphism

$$\varphi : \mathrm{Lie}(G) \oplus \mathrm{Lie}(G) \to T_e(G) \oplus T_e(H) \cong T_e(G \times H).$$

Therefore it suffices to show that $\varepsilon^{-1} \circ \varphi : \mathrm{Lie}(G \times H) \to \mathrm{Lie}(G) \oplus \mathrm{Lie}(H)$ is a Lie algebra homomorphism. For this we have

$$\begin{aligned}
(\varepsilon^{-1} \circ \varphi)[(X_1,X_2),(Y_1,Y_2)]_{(g,h)} &= (\varepsilon^{-1} \circ \varphi)([X_1,X_2],[Y_1,Y_2])_{(g,h)} \\
&= \varepsilon^{-1}([X_1,X_2]_e, Y_1,Y_2]_e)_{(g,h)} \\
&= \mathrm{d}(L_{(g,h)})_{(e,e)}([X_1,X_2] \oplus [Y_1,Y_2])_{(e,e)} \\
&= (L_{(g,h)})_{(e,e)}(X_1, \oplus X_2, Y_1 \oplus Y_2)_{(g,h)} \\
&= (X_1, \oplus X_2, Y_1 \oplus Y_2)_{(g,h)} \\
&= [(\varepsilon^{-1} \circ \varphi)(X_1,X_2), (\varepsilon^{-1} \circ \varphi)(Y_1,Y_2)]_{(g,h)}
\end{aligned}$$

so we have the desired Lie algebra isomorphism.

Problem 8-21. The one-dimensional Lie algbra must be abelian, isomorphic to $(\mathbb{R}, +)$. We may identify it with the subalgebra of $\mathfrak{gl}(2, \mathbb{R})$ consisting of matrices of the form

$$\begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix}, \qquad x \in \mathbb{R}.$$

Similarly, if the two-dimensional Lie algebra is abelian then it is isomorphic to the algebra of matrices

$$\begin{pmatrix} x_1 & 0 \\ 0 & x_2 \end{pmatrix}, \quad x_1, x_2 \in \mathbb{R}.$$

For the nonabelian case, choose a basis X, Y for the Lie algebra. Then $[X, Y] = aX + bY$ for some $a, b \in \mathbb{R}$. Assume that $b \neq 0$. Dividing by b gives

$$\left[\frac{1}{b}\, X,\ Y\right] = \underbrace{\frac{a}{b}X + Y}_{:=Z}.$$

Then we have

$$Z = \left[\frac{1}{b}\, X, Z - \frac{a}{b}X\right] = \left[\frac{1}{b}\, X, Z\right] = [W, Z]$$

where we called $W = b^{-1}X$. Therefore by relabeling we can take $[X, Y] = Y$ wlog. Assign the matrices

$$Y \longleftrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad X \longleftrightarrow \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

This determines an isomorphism into the subalgebra of matrices with the form

$$\begin{pmatrix} x_1 & 0 \\ x_2 & 2x_1 \end{pmatrix}, \qquad x_1, x_2 \in \mathbb{R}.$$

Remark: Note that this shows $\mathbb{S}^2$ is not a Lie group. To see this, first note that the connected Lie group corresponding to the abelian two-dimensional algebra is isomorphic to $\mathbb{R}^2$, which is not diffeomorphic to $\mathbb{S}^2$. For the non-abelian algebra, we calculate the generators

$$\exp tY = \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix}, \qquad \exp tX = \begin{pmatrix} t & 0 \\ t & t \end{pmatrix}.$$

The Lie group generated is clearly not compact, and hence not diffeomorphic to $\mathbb{S}^2$.

Problem 8-28. To compute $\mathfrak{sl}(n, \mathbb{C})$ first note $\det : GL(n, \mathbb{C}) \to \mathbb{C}^\times$ is a surjective Lie group homomorphism, and $SL(n, \mathbb{C}) = \ker(\det)$. It follows that $T_I SL(n, \mathbb{C}) = \ker \mathrm{d}(\det)_I$ where I is the $n \times n$ identity matrix. Let $M \in SL(n, \mathbb{C})$ and consider the path $\gamma : (-\varepsilon, \varepsilon) \to SL(n, \mathbb{C})$ defined for some $\varepsilon > 0$ by $\gamma(t) = I + Mt$. Then we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \det(\gamma(t)) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \sum_\sigma \operatorname{sgn}(\sigma) \cdot \gamma_1^{\sigma(1)} \dots \gamma_n^{\sigma(n)} = \operatorname{tr} M = 0$$

Where the sum is over permutations σ of $\{1, \dots, n\}$. The penultimate equality holds because the coefficient of the linear term in the sum is exactly the sum of the diagonal entries of M. Hence the Lie algebra is $\mathfrak{sl}(n, \mathbb{C}) = \{M \in \mathcal{M}(n, \mathbb{C}) : \operatorname{tr} M = 0\}$. Its complex dimension is $n^2 - 1$ because if $M_1^1, M_2^2, \dots M_{n-1}^{n-1}$ are given, M_n^n is determined by solving $\operatorname{tr} M = 0$.

Problem 8-29. 1. $\mathrm{Lie}(SL(n,\mathbb{R})) \simeq \mathfrak{sl}(n,\mathbb{R})$.

2. $\mathrm{Lie}(SO(n,\mathbb{R})) \simeq \mathfrak{o}(n,\mathbb{R})$. Define $\Phi : \mathfrak{gl}(n,\mathbb{R}) \longrightarrow \mathfrak{gl}(n,\mathbb{R}$ by $\Phi(A) = A^T A$. Clearly, $\Phi^{-1}(I) \cup \det^{-1}(1) = SO(n)$ and $\mathrm{d}\Phi(I) = A + A^T$. So, $\ker(\mathrm{d}\Phi(I)) = \mathfrak{o}(n)$, as required.

3. $\mathrm{Lie}(SL(n,\mathbb{C})) \simeq \mathfrak{sl}(n,\mathbb{C})$. The procedure is essentially the same as in 1.

4. $\mathrm{Lie}(U(n)) \simeq \mathfrak{u}(n)$. The procedure is very similar to 2. Define $\Phi : \mathfrak{gl}(n,\mathbb{C}) \longrightarrow \mathfrak{gl}(n,\mathbb{C})$ by $\Phi(A) = A^* A \Rightarrow \mathrm{d}\Phi(I) = A + A^*$. This yields:
$$\ker(\mathrm{d}\Phi(I)) = \{A \in \mathfrak{gl}(n,\mathbb{C}), A = -A^*\}.$$

5. $\mathrm{Lie}(SU(n)) \simeq \mathfrak{su}(n)$. Define Φ as in 4. So, the kernel is the same as before, with the added condition elements of the kernel have determinant 1 - yielding $\mathfrak{su}(n) = \mathfrak{u} \cap \mathfrak{sl}(n)$ - as required.

Problem 8-31. Define the bracket on $\mathfrak{g}/\mathfrak{h}$ by $[X + \mathfrak{h}, Y + \mathfrak{h}] = [X, Y] + \mathfrak{h}$. This is well defined since if $H_1, H_2 \in \mathfrak{h}$ then
$$[X + H_1, Y + H_2] = [X, Y] + [X, H_2] + [Y, H_1] + [H_1, H_2] \in [X, Y] + \mathfrak{h}$$
since $\mathfrak{h}$ is an ideal. Then the projection $\pi : \mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ is a homomorphism (trivial). Now if $\mathfrak{h}$ is an ideal we have just shown $\ker \pi = \mathfrak{h}$. Conversely, suppose $\varphi : \mathfrak{g} \to \mathfrak{v}$ is a homomorphism of Lie algebras. Let $Y \in \ker \varphi$ and $X \in \mathfrak{g}$. Then
$$\varphi([X, Y]) = [\varphi(X), \varphi(Y)] = [0, \varphi(Y)] = 0$$
so $[X, Y] \in \ker \varphi$. Hence $\ker \varphi$ is an ideal.

Problem 12-1. Let V and W be 2 dimensional vector spaces (over $\mathbb{R}$). Take linearly independent pairs $v_1, v_2 \in V$ and $w_1, w_2 \in W$. Then $v_1 \otimes w_1 + v_2 \otimes w_2$ is not expressible as a simple tensor, since if
$$v_1 \otimes w_1 + v_2 \otimes w_2 = v \otimes w$$
then we may write $v \otimes w$ in the basis $\{v_i \otimes w_j, \quad i, j = 1, 2\}$ of $V \otimes W$. But then
$$v \otimes w = (v_1 + v_2) \otimes (w_1 + w_2) = v_1 \otimes w_1 + v_2 \otimes w_2 + v_1 \otimes w_2 + v_2 \otimes w_1$$
a contradiction.

Problem 12-2. Define $\Phi : \mathbb{R} \times W \to W$ by scalar multiplication. Φ descends to a linear map $\widetilde{\Phi} : \mathbb{R} \otimes W \to W$ by the characteristic property of the tensor product, defined by the assigment $c \otimes v \to c \cdot v$. This is clearly a linear isomorphism since the kernel of $\widetilde{\Phi}$ is trivial and $1 \otimes v \to v$, so it is also surjective. Thus $\mathbb{R} \otimes V \cong V$. By the same argument $V \otimes R \cong V$.

Problem 12-3. Let (Z, B) have the universal property of the tensor product i.e. any bilinear map $\Phi : V \times W \to Y$ factors through Z. Consider the following commutative diagram:

In the triangle on the right, π factors through Z. On the left, B factors through $V \otimes W$ since it has the universal property as well. But then $\Phi \circ \Psi$ must be the identity on $V \otimes W$, so $\Psi = \Phi^{-1}$. Whence $\Phi : V \otimes W \to Z$ is a linear isomorphism.

$$\begin{tikzcd}
V \otimes W \arrow[r, "\Phi"] & Z \arrow[r, "\Psi"] & V \otimes W \\
& V \times W \arrow[ul, "\pi"] \arrow[u, "B"] \arrow[ur, "\pi"'] &
\end{tikzcd}$$

Problem 12-4. Define $\Phi : V_1^* \times \cdots \times V_k^* \times W \to L(V_1, \ldots, V_k; W)$ by

$$\Phi(\omega^1, \cdots, \omega^k, w)(v_1, \ldots v_k) = \omega^1(v_1)\omega^2(v_2)\ldots\omega^k(v_k) \cdot w$$

where $w \in W$. This assigment is linear in each ω^i,v_i and in w. By the characteristic property of the tensor product we get a unique (canonical) linear map $\widetilde{\Phi} : V_1^* \otimes \cdots \otimes V_k^* \otimes W \to L(V_1, \ldots, V_k; W)$. The standard basis for the tensor product is

$$\mathcal{B} = \left\{ \varepsilon_{(1)}^{i_1} \otimes \cdots \otimes \varepsilon_{(k)}^{i_k} \otimes w_j : 1 \leq i_1 \leq n_1, \ldots, 1 \leq i_k \leq n_k, 1 \leq j \leq n \right\}$$

as in Proposition 2.4, where $\{w_1, \ldots, w_j\}$ is a basis for W. $\mathcal{B}$ is mapped by $\widetilde{\Phi}$ to the basis of $L(V_1, V_2, \ldots, V_k; W)$.

Problem 12-7. If $e^1 \otimes e^2 \otimes e^3 = \alpha + \beta$ where $\alpha \in \Lambda^3(\mathbb{R}^*)$ and $\beta \in \Sigma^3(\mathbb{R}^*)$, then we must have that

$$(\alpha + \beta)(v_2, v_1, v_3) + (\alpha + \beta)(v_1, v_2, v_3) = 2\beta(v_1, v_2, v_3)$$

is symmetric for all $v_1, v_2, v_3 \in \mathbb{R}^3$. But $e^1 \otimes e^2 \otimes e^3$ does not have this property. e.g. take $v_i = e_i$ to be the standard basis so

$$e^1 \otimes e^2 \otimes e^3(e_1, e_2, e_3) + e^1 \otimes e^2 \otimes e^3(e_2, e_1, e_3) = 1.$$

On the other hand

$$e^1 \otimes e^2 \otimes e^3(e_1, e_3, e_2) + e^1 \otimes e^2 \otimes e^3(e_3, e_1, e_2) = 0.$$

Problem 12-8. Using the transformation rule for tangent vectors we have

$$\begin{aligned}
A_{i_1, \ldots i_k} &= A\left(\frac{\partial}{\partial x^{i_1}}, \ldots, \frac{\partial}{\partial x^{i_k}}\right) = A\left(\frac{\partial \tilde{x}^{j_1}}{\partial x^{i_1}}\frac{\partial}{\partial \tilde{x}^{j_1}}, \ldots, \frac{\partial \tilde{x}^{j_k}}{\partial x^{i_k}}\frac{\partial}{\partial \tilde{x}^{j_k}}\right) \\
&= \frac{\partial \tilde{x}^{j_1}}{\partial x^{i_1}} \cdots \frac{\partial \tilde{x}^{j_k}}{\partial x^{i_k}} A\left(\frac{\partial}{\partial \tilde{x}^{j_1}}, \ldots, \frac{\partial}{\partial \tilde{x}^{j_k}}\right) \\
&= \frac{\partial \tilde{x}^{j_1}}{\partial x^{i_1}} \cdots \frac{\partial \tilde{x}^{j_k}}{\partial x^{i_k}} A_{j_1 \ldots j_k}.
\end{aligned}$$

where the Einstein summation is over the k indices $j_1, j_2, \ldots j_k$.

Problem 13-.

Problem 13-21. Let (M, g) be a Riemannian manifold and $f \in C^\infty(M)$, and let $p \in M$ be a regular point of f.

(a) Let $v \in T_pM$, then

$$df_p(v) = \langle \operatorname{grad} f_p, v\rangle \leq |\operatorname{grad} f_p| \cdot |v|.$$

The inequality in Cauchy Schwarz is sharp iff v is in the direction of $\operatorname{grad} f_p$, and in that case $|v| = |\operatorname{grad} f_p|$.

(b) Let $S = f^{-1}(c)$, where $c = f(p)$ is a regular value. It follows that S is an embedded hypersurface. For any $q \in S$ and any tangent vector $v \in T_qS$, choose a path $\gamma : (-\epsilon, \epsilon) \to S$ such that $\gamma(0) = q$, $\dot\gamma(0) = v$. It follows that

$$\langle \operatorname{grad} f_q, v\rangle = df_q(v) = \frac{d}{dt}\Big|_{t=0} f(\gamma(t)) = 0.$$

Hence $\operatorname{grad} f$ is orthogonal to S.

Problem 13-23. To find a covector field on $\mathbb{S}^2$ that vanishes at exactly one point, just take the vector field defined in problem $8-13$ and consider the dual covector field defined by some Riemannian metric.

Problem 13-24. Let M be a compact smooth manifold and $f : M \to \mathbb{R}$ a smooth function with critical points $p_1, \ldots, p_k$ with respective critical values $c_1, \ldots, c_k$, labeled so that $c_1 \leq c_2 \leq \cdots \leq c_k$. For any $a < b$ define $M_a = f^{-1}(a)$, $M_{[a,b]}) = f^{-1}([a,b])$, and $M_{(a,b)} = f^{-1}((a,b))$. If a and b are regular values, M_a and M_b are embedded hypersurfaces in M, $M_{(a,b)}$ is an open submanifold, and $M_{[a,b]}$ is a regular domain (a properly embedded codimension 0 submanifold with boundary).

(a) Define $X = \operatorname{grad} f/|\operatorname{grad} f|^2$ on $M \setminus \{p_1, \ldots, p_k\}$ with respect to some Riemannian metric on M. Let θ be the flow of X. Then $f(\theta_t(p)) = f(p) + t$ whenever $\theta_t(p)$ is defined. To see this, we note that

$$\frac{d}{dt}\Big|_{t=0} f(\theta_t(p)) = \langle \operatorname{grad} f_p, X_p\rangle = 1.$$

so $f(\theta_t(p))$ is a solution to the ODE $\gamma(0) = f(p)$, $\dot\gamma(0) = 1$. Hence by uniqueness the equality holds.

(b) Suppose $[a,b]$ contains no critical values of f. Then we claim θ restricts to a diffeomorphism from $[0, b-a] \times M_a$ to $M_{[a,b]}$. This follows from (a) since

Problem 14-1. If $\omega^1, \ldots, \omega^k$ is linearly dependent, extend them to a basis $\omega^1 \ldots \omega^n$ for $\Omega^1(M)$ and write $\omega = c_i\, \omega^i$ where the summation is over $i = 2, \ldots, n$. Then

$$\omega^1 \wedge \cdots \wedge \omega^k = (c_i \cdot \omega^i) \wedge \omega^2 \cdots \wedge \omega^k = c_1\, \omega^2 \wedge \omega^2 \cdots \wedge \omega^k + \cdots + c_k\, \omega^k \wedge \cdots \wedge \omega^k = 0.$$

To show the converse suppose $\omega^1 \wedge \cdots \wedge \omega^k \neq 0$. Now $\omega^1, \ldots, \omega^k$ must be linearly independent, for otherwise the above caluculation would show that the wedge product is 0.

$$\begin{tikzcd} V^* \otimes \cdots \otimes V^* \arrow[r, "\sim"] \arrow[d, "\pi"] & T^k(V^*) \arrow[d, "\mathrm{Alt}"] \\ A^k(V^*) \arrow[r, "F"] & \Lambda^k(V^*) \end{tikzcd}$$

Problem 14-3. For an equivalence class $[\omega^1 \otimes \cdots \otimes \omega^k] \in V^* \otimes \cdots \otimes V^*/\mathcal{A}$ we must define

$$F[\omega^1 \otimes \cdots \otimes \omega^k] = \sum_{\sigma \in S_k} (\mathrm{sgn}\sigma)\omega^{\sigma(1)} \otimes \cdots \otimes \omega^{\sigma(k)}$$

in order that the diagram commutes:

F is independent of the choice of equivalence class representative since any element of $\mathcal{A}$ (i.e. with $\omega^i = \omega^j$) is annihilated in the sum. The exterior product on $\bigoplus_k A^k(V^*)$ is defined by $\omega \wedge \eta = \pi(\tilde{\omega} \otimes \tilde{\eta})$ where $\tilde{\eta} \in \pi^{-1}(\eta)$ and $\tilde{\omega} \in \pi^{-1}(\omega)$. This is independent of the choice of $\tilde{\omega}$ and $\tilde{\eta}$ since for any $\alpha_1, \alpha_2 \in \mathcal{A}$

$$(\tilde{\omega} + \alpha_1) \otimes (\tilde{\eta} + \alpha_2) = \tilde{\omega} \otimes \tilde{\eta} + \alpha_1 \otimes \tilde{\eta} + \tilde{\omega} \otimes \alpha_2 + \alpha_1 \otimes \alpha_2 \sim \tilde{\omega} \otimes \tilde{\eta}$$

where $\sim$ indicates equivalence under the quotient. We also have$F(\omega \wedge \eta) = \mathrm{Alt}(\omega \otimes \eta)$.

Problem 14-5. We must show that $\alpha^i \in \mathrm{span}\{\omega^1, \ldots, \omega^k\}$ for each $i = 1, \ldots, k$. Arguing to the contrary, suppose $\alpha^1, \omega^1, \ldots, \omega^k$ are linearly independent on U. Then

$$0 = \left(\sum_{i=1}^{k} \alpha^i \wedge \omega^i \right) \wedge (\omega^2 \wedge \cdots \wedge \omega^k) = \alpha^1 \wedge \omega^1 \wedge \omega^2 \wedge \cdots \wedge \omega^k$$

where all the terms with repeated terms vanish. But $\alpha^1, \omega^1, \ldots, \omega^k$ linearly independent implies $\alpha^1 \wedge \omega^1 \wedge \cdots \wedge \omega^k \neq 0$, a contradiction. Hence each α^i is in the span of the ω^j's.

Problem 14-7. (a) $\mathrm{d}\omega = \mathrm{d}x \wedge \mathrm{d}y$. $F^*\omega = st\ \mathrm{d}(e^t) = ste^t\ \mathrm{d}t$. Hence

$$\begin{aligned} \mathrm{d}F^*\omega &= te^t\ \mathrm{d}s \wedge \mathrm{d}t \\ F^*\mathrm{d}\omega &= \mathrm{d}(st) \wedge \mathrm{d}(e^t) = (t\mathrm{d}s + s\mathrm{d}t) \wedge e^t\ \mathrm{d}t = te^t\ \mathrm{d}s \wedge \mathrm{d}t. \end{aligned}$$

(b)

$$\begin{aligned} F^*\omega &= u^2(1-u^2-v^2)^{-1/2}\ \mathrm{d}v \wedge \mathrm{d}u + \ v^2(1-u^2-v^2)^{-1/2}\mathrm{d}v \wedge \mathrm{d}u + (1-u^2-v^2)^{1/2}\mathrm{d}u \wedge \mathrm{d}v \\ &= (1-u^2-v^2)^{-1/2}(1-2u^2-2v^2)\mathrm{d}u \wedge \mathrm{d}v \end{aligned}$$

Problem 14-9. Assume that the fibre $\pi^{-1}(\{p\})$ is a connected submanifold of M for all $p \in N$. Let $\omega \in \Omega^k(M)$ such that $v \lrcorner\, \omega_p = v \lrcorner\, \mathrm{d}\omega_p = 0$ for all vertical vectors v (i.e. $\mathrm{d}\pi(v) = 0$). For vector fields $V_1, \ldots, V_k$ Let

$$\eta(\pi_* V_1, \ldots, \pi_* V_k) = \omega(V_1, \ldots, V_k).$$

We claim that η is well-defined. First note that if $\pi_* V_1 = \pi_* V_1'$ then $\omega(V_1, \dots, V_k) = \omega(V_1', \dots V_k)$ by the hypothesis that π annihilates vertical vectors. Now if the pushforward of $X \in \mathfrak{X}(M)$ exists and $Y \in \ker \mathrm{d}\pi$ then $[X, Y] \in \ker \mathrm{d}\pi$. From the product formula we have

$$Y(\omega(V_1, \dots, V_k)) = (\mathcal{L}_Y \omega)(V_1, \dots, V_k) + \omega([Y, X_1], X_2, \dots, X_k) + \omega(X_1, \dots, [Y, X_k]) = 0$$

where the Lie derivative is 0 since $Y \lrcorner \mathrm{d}\omega = 0$. Hence ω is constant on the fibres, so η is well-defined. The converse is obvious since $v \lrcorner \omega = v \lrcorner \pi^* \eta = 0$ and $v \lrcorner \mathrm{d}\omega = v \lrcorner \pi^*(\mathrm{d}\eta) = 0$

Problem 15-1. Suppose $M = U \cup V$ where U and V are orientable open submanifolds and $U \cap V$ is connected. Choose orientations for $\mathcal{O}$ and $\mathcal{O}'$ for U and V respectively. Then at $p \in U \cap V$ we must have $\mathcal{O}_p = \mathcal{O}'_p$ by Proposition 15.9 since $U \cap V$ is a connected submanifold. By the same proposition the orientations on U and V are equal and so determine an orientation of M whose restriction to U (resp. V) is $\mathcal{O}$ (resp. $\mathcal{O}'$).

Problem 15-3. Consider the closed unit ball $\mathbb{B}^n \subset \mathbb{R}^n$ with the induced flat metric. Let N be the unit outward pointing vector field along $\mathbb{S}^{n-1}$ given by $V(x) = x$. By Corollary 15.34 the volume form for $\mathbb{S}^{n-1}$ is given by

$$\omega = \iota^*_{\mathbb{S}^{n-1}}(N \lrcorner \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n) = \sum_{i=1}^{n} (-1)^{i-1} x^i \, \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \dots \mathrm{d}x^n.$$

α is orientation preserving iff $\alpha^* \omega = c\omega$ for some $c > 0$. But $\alpha^* \omega = (-1)^n \omega$ so this is true iff $n - 1$ is odd.

Problem 15-5. It suffices to show that TM and T^*M have consistently oriented atlases. Let $(U, \varphi), (V, \psi)$ be charts for $p \in M$ and (x^i) coordinate functions for φ.
We get the corresponding tangent-bundle charts $(\pi^{-1}(U), \tilde{\varphi}), (\pi^{-1}(V), \tilde{\psi})$ where $\pi : TM \to M$ is the projection map and the $(\tilde{x}^i)$ are tangent-lifted coordinates for φ (as in Proposition 3.18). The determinant of the transition map $\tilde{\psi} \circ \tilde{\varphi^{-1}}$ is simply the determinant of the matrix

$$\begin{pmatrix} \dfrac{\partial \tilde{x}^i}{\partial x^j}(p) & 0 \\ \cdot & \dfrac{\partial \tilde{x}^i}{\partial x^j}(p) \end{pmatrix}$$

which is a square and hence positive. An identical argument works for T^*M with the cotangent charts as in Proposition 11.9.

Problem 16-5. Let $H : M \times [0, 1] \to N$ be a homotopy between F and G. By Whitney's approximation theorem we make take H to be a smooth homotopy. Take the standard orientation on $[0, 1]$ and equip $M \times [0, 1]$ with the product orientation. Let η be an orientation form for N. By Stokes's theorem,

$$\begin{aligned} 0 &= \int_{M \times [0,1]} H^* \mathrm{d}\eta = \int_{M \times [0,1]} \mathrm{d}H^* \eta \\ &= \int_{\partial M \times \{0,1\}} H^* \eta = \int_M F^* \eta - \int_M G^* \eta. \end{aligned}$$

Therefore by Proposition 16.6 (d), F and G must both be orientation preserving or orientation reversing.

Problem 16-6. (**Hairy Ball Theorem**)
The following are equivalent:

(a) There exists a non-vanishing vector field on $\mathbb{S}^n$.

(b) There exists a continuous map $V : \mathbb{S}^n \to \mathbb{S}^n$ satisfying $V(x) \perp x$ for all $x \in \mathbb{S}^n$.

(c) The antipodal map $\alpha : \mathbb{S}^n \to \mathbb{S}^n$ is homotopic to $\mathrm{id}_{\mathbb{S}^n}$.

(d) The antipodal map $\alpha : \mathbb{S}^n \to \mathbb{S}^n$ is orientation-preserving.

(e) n is odd.

Proof. $(a) \implies (b)$: If X is a nowhere-vanishing vector field on $\mathbb{S}^n$ then define $V(x) = X(x)/|X(x)|$ where $|\cdot|$ is the Euclidean norm.

$(b) \implies (c)$: Define the homotopy $H : \mathbb{S}^n \times [0,1] \to \mathbb{S}^n$ between V and α by $H(x,t) = tV(x) - \sqrt{1-t^2}\, x$. By orthogonality $|H(x,t)| = 1$ for each t so H is well-defined. There is a similarly defined homotopy between V and $\mathrm{Id}_{\mathbb{S}^n}$. Since begin homotpic is an equivalence relation, $\alpha \simeq \mathrm{Id}_{\mathbb{S}^n}$.

$(c) \implies (d)$: it follows from problem 16-5 that α is orientation preserving since the identity is.

$(d) \implies (e)$: Problem 15-3.

$(e) \implies (a)$: For n odd we view $\mathbb{S}^n$ as being embedded in $\mathbb{C}^{n+1}$. Define the flow $\theta(t,z) = e^{it}z$. The infintesimal generator is

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \theta^{(z)}(t) = iz$$

which defines a smooth non-vanishing vector field on $\mathbb{S}^n$. $\square$

Problem 16-11. We'll cheat and use $\operatorname{div} X = \star \mathrm{d} \star X^\flat$. It follows immediately from problem 16-21 that

$$\operatorname{div}\left(X^i \frac{\partial}{\partial x^i}\right) = \sqrt{g^{-1}} \frac{\partial}{\partial x^i}(X^i \sqrt{g})$$

where $g = \det g_{ij}$ and $g^{-1} = \det g^{ij}$.

Problem 16-12. Using notation from 6-11, the product rule gives us

$$\operatorname{div}\left(f X^i \frac{\partial}{\partial x^i}\right) = f \operatorname{div} X + X^i \frac{\partial f}{\partial x^i} = f \operatorname{div} + \langle \operatorname{grad} f, X\rangle_g.$$

Thus if (M,g) is compact we can integrate and use the divergence theorem:

$$\int_M \operatorname{grad}\langle f, X\rangle_g \mathrm{d}V_g = \int_{\partial M} f\langle X, N\rangle_g \mathrm{d}V_{\tilde{g}} - \int_M (f \operatorname{div}) \mathrm{d}V_g \tag{1}$$

where N is the unit outward pointing vector field and $\widetilde{g}$ is the metric induced on ∂M. In the Euclidean case this is familiar integration by parts

$$\int_{[a,b]} f'g \, \mathrm{d}x = fg\Big|_a^b - \int_{[a,b]} fg' \mathrm{d}x.$$

Problem 16-13. Define the "geometer's Laplacian" by $\Delta u = -\mathrm{div}\ (\mathrm{grad}\ u)$. Using problem 6-11 and the formula for the gradient we get

$$\begin{aligned} -\operatorname{div}(\operatorname{grad} u) &= -\operatorname{div}\left(g^{ij}\frac{\partial u}{\partial x^i}\frac{\partial}{\partial x^j}\right) \\ &= -\sqrt{g^{-1}}\frac{\partial}{\partial x^i}\left(g^{ij}\sqrt{g}\frac{\partial u}{\partial x^j}\right). \end{aligned}$$

On $\mathbb{R}^n$ with the Euclidean metric this reduces to

$$\begin{aligned} \Delta u &= -\frac{\partial}{\partial x^i}\left(\delta^i_j\frac{\partial u}{\partial x^j}\right) \\ &= -\sum_{i=1}^n \frac{\partial^2 u}{(\partial x^i)^2}. \end{aligned}$$

Problem 16-14. (a) Using (1) with $X = \operatorname{grad} v$ and $f = u$ we obtain

$$\int_M u\Delta v \, \mathrm{d}V_g = \int_M \langle \operatorname{grad} u, \operatorname{grad} v\rangle_g \mathrm{d}V_g - \int_{\partial M} uNv \, \mathrm{d}V_{\widetilde{g}} \tag{2}$$

since by definition $\langle \operatorname{grad} v, N\rangle_g = \mathrm{d}v(N)$. Interchanging the role of u and v and subtracting the resulting equation from (2), we get

$$\int_M u\Delta v - v\Delta u \mathrm{d}V_g = \int_{\partial M} (vNu - uNv) \, \mathrm{d}V_{\widetilde{g}}.$$

(b) Now suppose M has no boundary and v is harmonic. Then (2) becomes

$$\int_M \langle \operatorname{grad} u, \operatorname{grad} v\rangle_g \mathrm{d}V_g = 0.$$

By Proposition 16.28 this is equivalent to $\langle \operatorname{grad} u, \operatorname{grad} v\rangle_g = \mathrm{d}v(\operatorname{grad} u) \equiv 0$. Since $u \in C^\infty(M)$ is arbitrary we must have $\mathrm{d}v = 0$ so v is constant since M is connected.

(c) Now suppose $\partial M \neq \emptyset$ and let u, v be harmonic functions such that $u|_{\partial M} = v|_{\partial M}$. Since $u - v$ is again harmonic we have

$$0 = \int_{\partial M} ((u-v)N(u-v)) \, \mathrm{d}V_{\widetilde{g}} = \int_M \langle \operatorname{grad}(u-v), \operatorname{grad}(u-v)\rangle_g \mathrm{d}V_g.$$

By the same reasoning $u - v$ is constant on M. But since they agree on ∂M they must agree everywhere.

Problem 16-15. (a) Any harmonic function is an eigenfunction for 0, in this case the constant functions.

(b) Now suppose λ is a nonzero eigenvalue for Δ. From (2) we have

$$\int_M u^2 \mathrm{d}V_g = \lambda \int_M u\Delta u \mathrm{d}V_g = \int_M \langle \operatorname{grad} u, \operatorname{grad} u\rangle_g \, \mathrm{d}V_g \geq 0.$$

Thus $\lambda > 0$.

(c) Let u, v be eigenfunctions for λ_1, λ_2 respectively. Then

$$\begin{aligned}\int_M uv \, \mathrm{d}V_g &= \lambda_2 \int_M u\Delta v \, \mathrm{d}V_g \\ &= \lambda_2 \int_M \langle \operatorname{grad} u, \operatorname{grad} v\rangle_g \, \mathrm{d}V_g\end{aligned}$$

By the same token

$$\int_M uv \, \mathrm{d}V_g = \lambda_1 \int_M \langle \operatorname{grad} u, \operatorname{grad} v\rangle_g \, \mathrm{d}V_g.$$

Therefore if $\lambda_1 \neq \lambda_2$ the integral must vanish.

Remark: We can interpret the integral as the inner product (u, v) of 0-forms (problem 16-22).

Problem 16-18. (a) Define the inner product on $\Gamma^k(T_p^*M)$ for each $k = 1, \ldots, n$ by

$$\langle \varepsilon^I, \varepsilon^J\rangle = \det\langle (\varepsilon^{I_i})^\#, (\varepsilon^{J_j})^\#\rangle = \delta^I_J = \begin{cases} \operatorname{sgn}\sigma, & I = J_\sigma \text{ for some } \sigma \in S_n \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where (ε^i) is the coframe dual to a local orthonormal frame and I, J are increasing k-multi indices. This definition is independent of the orthonormal coframe since the change of basis matrix P is orthogonal and hence changes the determinant in (3) by a factor of $\det(P)\det(P^T) = 1$. Thus the inner product can be defined uniqely on the basis and extended by linearity.

(b) Clearly $\langle \mathrm{d}V_g, \mathrm{d}V_g\rangle = 1$ and $\mathrm{d}V_g$ is the unique orientation form with this property since any positively oriented n-form has $\langle \omega_p, \omega_p\rangle = c^2$ for some $c > 0$.

(c) Uniqueness of $\star$ follows from uniqueness of the inner product. Now for any k-multi-index $(i_1, \ldots, i_k)$ define $\star : \Gamma^k(T^*M) \to \Gamma^{n-1}(T^*M)$ by

$$\star(\varepsilon^{i_1} \wedge \cdots \wedge \varepsilon^{i_k}) = (\operatorname{sgn}\sigma)\varepsilon^{j_1} \wedge \cdots \wedge \varepsilon^{j_{n-k}}$$

where $(i_1, \ldots, i_k, j_1, \ldots, j_{n-k})$ is a permutation of $(1, \ldots, n)$ and $\sigma \in S_n$ is the permutation taking the former to the latter. From this definition we have

$$\varepsilon^I \wedge \star\varepsilon^J = (\operatorname{sgn}\sigma)\varepsilon^1 \wedge \cdots \wedge \varepsilon^n = \langle \varepsilon^I, \varepsilon^J\rangle \mathrm{d}V_g$$

whenever I and $(1, \ldots, n) \setminus J$ are disjoint. Otherwise both sides are 0.

(d) $\star$ on 0-forms is defined to be regular multiplication so this is obvious since $\star 1 = \mathrm{d}V_g$.

(e) It suffices to check this on a basis. Note that $\star\varepsilon^I \wedge \star\star\varepsilon^I = \mathrm{d}V_g = \varepsilon^I \wedge \star\varepsilon^I$. $\star\star\varepsilon^I$ is a k-form while $\star\varepsilon^I$ is an $(n-k)$-form so

$$(-1)^{k(n-k)} \star\star\varepsilon^I \wedge \star\varepsilon^I = \varepsilon^I \wedge \star\varepsilon^I.$$

which proves $\star^2 = (-1)^{k(n-k)}$.

Problem 16-21. Recall the above diagram for a 3-dimensional oriented Riemannian manifold (M, g).

$$\begin{tikzcd}
C^\infty(M) \arrow[r, "\mathrm{grad}"] \arrow[d] & \mathfrak{X}(M) \arrow[r, "\mathrm{curl}"] \arrow[d, "\flat"] & \mathfrak{X}(M) \arrow[r, "\mathrm{div}"] \arrow[d, "\beta"] & C^\infty(M) \arrow[d, "\star"] \\
\Omega^0(M) \arrow[r, "\mathrm{d}"] & \Omega^1(M) \arrow[r, "\mathrm{d}"] & \Omega^2(M) \arrow[r, "\mathrm{d}"] & \Omega^3(M)
\end{tikzcd}$$

First we prove $\beta(X) = \star X^\flat$. First notice that in local coordinates (x^i) we have $\mathrm{d}x^i \wedge \star\mathrm{d}x^j = g^{ij}\ \mathrm{d}V_g$. Thus by comparing coefficients $\star\mathrm{d}x^i = g^{ij}\sqrt{g}\ \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \ldots \mathrm{d}x^n$. Suppose $X = X^i\partial_i$, then $X^\flat = X_i\ \mathrm{d}x^i$ and

$$\begin{aligned}
\star X^\flat &= \sum_{i=1}^n (-1)^{i-1} X_j g^{ij} \sqrt{g}\ \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \mathrm{d}x^n \\
&= \sum_{i=1}^n (-1)^{i-1} X^i \sqrt{g}\ \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \mathrm{d}x^n \\
&= X \lrcorner\, \mathrm{d}V_g.
\end{aligned}$$

By the diagram $\operatorname{div} X = \star^{-1}\mathrm{d}\beta(X) = \star\mathrm{d}\star X^\flat$. Similarly

$$\operatorname{curl} X = \beta^{-1}(\mathrm{d}X^\flat) = (\star^{-1}\mathrm{d}X^\flat)^\sharp = (\star\mathrm{d}X^\flat)^\sharp.$$

Problem 16-22. (a) Let (M, g) be an n-dimensional compact oriented Riemmanian manifold. If $\omega \in \Omega^k(M)$ we have

$$\mathrm{d}^* \circ \mathrm{d}^*\ \omega = (-1)^{n(k+1)+1} \star\mathrm{d}\star\star\mathrm{d}\star\omega = (-1)^{n(k+1)+1} \cdot (-1)^{(n-k+1)(k-1)} \star\mathrm{d}\circ\mathrm{d}\star\omega = 0.$$

(b) Define the L^2 inner product on p-forms

$$(\omega, \eta) = \int_M \langle \omega, \eta \rangle_g\ \mathrm{d}V_g.$$

This is indeed an inner product since $\langle \cdot, \cdot \rangle$ is an inner product and the integral is linear.

(c) Let ω be a k- form and η be a $(k-1)$- form. We claim d^* is the "adjoint" of d i.e. $(\mathrm{d}^*\omega, \eta) = (\omega, \mathrm{d}\eta)$. Using the product rule for the exterior derivative, the RHS is

$$\begin{aligned}(\mathrm{d}\eta, \omega) &= \int_M \mathrm{d}\eta \wedge \star\omega \\ &= \int_M \mathrm{d}(\eta \wedge \star\omega) - (-1)^{k-1}\eta \wedge \mathrm{d} \star \omega \\ &= \int_M (-1)^k \eta \wedge \mathrm{d} \star \omega.\end{aligned}$$

On the other hand the LHS is

$$\begin{aligned}((-1)^{n(k+1)+1} \star \mathrm{d} \star \omega, \eta) &= \int_M (-1)^{n(k+1)+1}\ \eta \wedge \star\star \mathrm{d} \star \omega \\ &= \int_M (-1)^{n(k+1)+1} \cdot (-1)^{(n-k+1)(k-1)}\ \eta \wedge \mathrm{d} \star \omega.\end{aligned}$$

Finally $n(k+1) + 1 + (n-k+1)(k-1) = 2nk + 2k - k^2 \equiv k \pmod 2$, so we get equality.

Problem 17-1. If $\theta \in \Omega^{p-1}(M)$ and $\xi \in \Omega^{q-1}(M)$ then

$$(\omega + \mathrm{d}\theta) \wedge (\eta + \mathrm{d}\xi) = \omega \wedge \eta + \mathrm{d}\theta \wedge \xi + \omega \wedge \mathrm{d}\xi + \mathrm{d}\theta \wedge \mathrm{d}\xi.$$

The last three terms are clearly closed, which shows that the cup product $[\omega] \cup [\eta] = [\omega \wedge \eta]$ is well-defined.

Problem 17-2. The Laplace Beltrami operator on differential forms $\Delta : \Omega^p(M) \to \Omega^p(M)$ is defined by $\Delta = \mathrm{d} \circ \mathrm{d}^* + \mathrm{d}^* \circ \mathrm{d}$.
$(a) \iff (b)$: trivial
$(b) \iff (c)$: Let $||\cdot||$ denote the norm induced by the inner product $(\cdot, \cdot)$. We must show that if $\mathrm{d}\omega = 0$ and $\mathrm{d}^*\omega = 0$ then ω is the unique form that minimizes the norm over the cohomology class $[\omega]$. First we prove uniqueness. Suppose ω and η are two cohomologous harmonic p-forms. Then

$$(\omega_1 - \omega_2, \omega_1 - \omega_2) = (\omega_1 - \omega_2, \mathrm{d}\theta) = (d^*\omega - d^*\omega, \theta) = 0.$$

for some $\theta \in \Omega^{p-1}(M)$, so $\omega_1 = \omega_2$. To show ω is a stationary point of the norm functional we consider a variation $f(t) = ||\omega + \mathrm{d}(t\mathrm{d}^*\omega)||^2$. Then we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} f(t) = 2(\omega, \mathrm{d}\mathrm{d}^*\omega) = 2(\mathrm{d}^*\omega, \mathrm{d}^*\omega) = 2||\mathrm{d}^*\omega||^2 = 0.$$

As the norm is bounded below we conclude that ω is a minimum. Conversely note that the derivative at $t = 0$ vanishes only if $d^*\omega = 0$.

Problem 17-5. Choose an open cover $\{U, V\}$ such that U contains $-e_1$ but not e_1 and similarly V contains e_1 but not $-e_1$. Then the Mayer Vietoris sequence is

$$\cdots \to H^{p-1}_{dR}(\mathbb{R}^n) \to H^p_{dR}(\mathbb{R}^n \backslash \{\pm e_1\}) \to H^p_{dR}(\mathbb{R}^n \backslash \{e_1\}) \oplus H^p_{dR}(\mathbb{R}^n \backslash \{-e_1\}) \to H^p_{dR}(\mathbb{R}^n) \to \cdots .$$

If $p > 1$ then both ends are 0 yeilding an isomorphism since

$$H^p_{dR}(\mathbb{R}^n \setminus \{e_1\}) \oplus H^p_{dR}(\mathbb{R}^n \setminus \{-e_1\}) \cong H^p_{dR}(\mathbb{S}^{n-1}) \oplus H^p_{dR}(\mathbb{S}^{n-1}) = \begin{cases} \mathbb{R}^2, & p = 0, n-1 \\ 0, & \text{otherwise} \end{cases}$$

For $p = 0$ we have $H^0_{dR}(\mathbb{R}^n \setminus \{\pm e_1\} = \mathbb{R}$ since it is connected. For $p = 1$ the bottom of the sequence reads

$$\cdots \to \underbrace{H^0_{dR}(U) \oplus H^0_{dR}(V)}_{\cong \mathbb{R}^2} \xrightarrow{i^* - j^*} \underbrace{H^0_{dR}(\mathbb{R}^n)}_{\cong \mathbb{R}} \to H^1_{dR}(\mathbb{R}^n \setminus \{\pm e_1\}) \to 0 \to \cdots$$

$i^* - j^*$ is surjective so the connecting homomorphism is injective. Since the sequence ends in 0 it is also surjective.

Problem 17-6. First suppose $p \neq 0, 1, n-1$. Choose a coordinate neighborhood U of x. Then $(M \setminus \{x\}) \cap U$ is homotopic to $\mathbb{S}^{n-1}$, so the Mayer Vietoris exact sequence is

$$\cdots \to 0 \to H^p_{dR}(M) \to H^p_{dR}(M \setminus \{x\}) \to 0 \to \cdots$$

since $H^p(\mathbb{S}^{n-1})$ is trivial for $p \neq n-1, 1$. The case $p = 0$ is trivial since both M and $M \setminus \{x\}$ are connected and hence have 0th degree cohomology isomorphic to $\mathbb{R}$. For $p = 1$, we have the exact sequence

$$\underbrace{H^0_{dR}(M)}_{\cong \mathbb{R}} \to \underbrace{H^0_{dR}(M \setminus \{x\}) \oplus H^0_{dR}(U)}_{\cong \mathbb{R}^2} \to \underbrace{H^0_{dR}(\mathbb{S}^{n-1})}_{\cong \mathbb{R}} \to H^1_{dR}(M) \xrightarrow{\iota^*} H^1_{dR}(M \backslash \{x\}) \to 0 \to \cdots .$$

Surjectivity is immediate, so it suffices to prove injectivity. Note that the second map in the sequence $\mathbb{R}^2 \to \mathbb{R}$ is surjective, so by exactness at $H^1_{dR}(M)$ the third map is trivial. It follows that ι^* is injective and hence an isomorphism.
Now suppose in addition M is compact and orientable. For the case $p = n-1$ first note that $M \setminus \{x\}$ is not compact so its top degree cohomology is trivial while on the other hand $H^n_{dR}(M) \cong \mathbb{R}$ (Theorem 17.31). Consequently we get the exact sequence

$$\cdots \to 0 \to H^{n-1}_{dR}(M) \xrightarrow{\iota^*} H^{n-1}_{dR}(M \setminus \{x\}) \to \underbrace{H^{n-1}_{dR}(\mathbb{S}^{n-1})}_{\cong \mathbb{R}} \to \underbrace{H^n_{dR}(M)}_{\cong \mathbb{R}} \to 0 \tag{4}$$

Which gives us injectivity. It remains is to show surjectivity. Since the penultimate map is an isomorphism, the image of the preceding map is trivial. Thus ι^* is surjective.

Problem 17-7. We may choose an open cover $\widetilde{M_1}, \widetilde{M_2}$ of $M_1 \# M_2$ such that $\widetilde{M_1} \approx M_1 \backslash \{p_1\}$, $\widetilde{M_2} \approx M_2 \setminus \{p_2\}$ for some $p_1, p_2 \in M_1 \# M_2$ and such that $\widetilde{M_1} \cap \widetilde{M_2} \approx (-1, 1) \times \mathbb{S}^{n-1}$. By

Problem 17-6 the inclusions $M_1 \setminus \{p_1\} \hookrightarrow M_1$, $M_2 \setminus \{p_2\} \hookrightarrow M_2$ induce isomorphisms on the de Rham cohomology groups for $0 < p < n-1$. Hence the Mayer Vietoris sequence is

$$\cdots \to H^{p-1}_{dR}((-1,1)\times\mathbb{S}^{n-1}) \to H^p_{dR}(M_1\#M_2) \to H^p_{dR}(M_1)\oplus H^p_{dR}(M_1) \to H^p_{dR}((-1,1)\times\mathbb{S}^{n-1}) \to \cdots$$

Note that $(-1,1)\times\mathbb{S}^{n-1}$ is homotopy equivalent to $\mathbb{S}^{n-1}$ since $(-1,1)$ is contractible. Thus the sequence yeilds an isomorphism for $p \neq 0, n-1$.
If M_1, M_2 are in addition compact and orientable then the second part of problem 17-6 shows the inclusions again induce isomorphisms. Since $M_1\#M_2$ is also compact and orientable its top cohomology is one-dimensional. So we obtain the sequence

$$\cdots 0 \to H^{n-1}_{dR}(M_1\#M_2) \to H^{n-1}_{dR}(M_1)\oplus H^{n-1}_{dR}(M_2) \to \mathbb{R} \to \mathbb{R} \to 0.$$

By the same reasoning as with (4) we see that this sequence also yeilds an isomorphism $H^{n-1}_{dR}(M_1\#M_2) \cong H^{n-1}_{dR}(M_1)\oplus H^{n-1}_{dR}(M_2)$.

Problem 17-8. If $\mathcal{O}$ is an orientation for M let ω be a corresponding orientation form. $H^n_{dR}(M)$ is one dimensional so it has a unique orientation where $[\omega]$ is a positively oriented basis. Conversely if $[\omega]$ is a positively oriented basis for $H^n_{dR}(M)$ then it determines an orientation on M for which ω is an orientation form. This orientation is independent of the choice of cohomology class representation since $H^n_{dR}(M)$ is spanned by any orientation form.

Let η be an orientation form for N. Then $F : M \to N$ is orientation preserving iff $F^*\eta$ is a positive multiple of ω. This is possible iff the induced map $F^* : H^n_{dR}(N) \to H^n_{dR}(M)$ is orientation preserving, i.e. $F^*[\eta] = c[\omega]$ for $c > 0$.

Problem 17-12. We prove the contrapositive: if F is not surjective then $\int_M F^*\eta = 0$ for all $\eta \in \Omega^n(N)$. Indeed, the degree of F is 0 since we may choose any $p \in N \setminus F(M)$, which is a regular value of F. Using the formula of Theorem 17.35 (b) shows $\deg F = 0$ since $F^{-1}(\{p\}) = \emptyset$.

To show that the converse is not necessarily true, we must find compact, orientable, connected manifolds M, N and a smooth surjective map $F : M \to N$ with degree 0. It thus suffices to choose F to be null-homotopic, so that the result follows from homotopy invariance of degree. For example, take $F : \mathbb{S}^1 \to \mathbb{S}^1$ given by $x + iy \mapsto e^{2\pi i x}$. F is clearly surjective and $H : \mathbb{S}\times[0,1] \to \mathbb{S}$, $(x+iy, t) \mapsto e^{2\pi i t x}$ is a null-homotopy.

Problem 17-13. It is convenient to view the torus $\mathbb{T}^2$ as the quotient space $\mathbb{R}^2/\mathbb{Z}^2$. Thus any element of $M_2(\mathbb{Z})$ acts on $\mathbb{T}^2$ since it preserves the integer lattice. From this point of view f corresponds to the 2×2 identity and g to the matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ which acts on $(z,w) = (e^{i\theta_1}, e^{i\theta_2})$ by matrix multiplication

$$\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} \theta_2 \\ -\theta_1 \end{pmatrix}.$$

The degree is simply the determinant, so f and g have the same degree. To show f and g are not homotopic, consider the induced homomorphisms f_* and g_* on $\pi_1(\mathbb{T}^2) = \mathbb{Z}^2$ (we do not reference a base point since the torus is path connected). f_* is the identity but g_* is the map $(n,m) \mapsto (m,-n)$. Since f_* and g_* are different, f and g cannot be homotopic.

Problem 18-2. We use the fact that the map $\varphi : \pi_1(M,q)$ sending $[\gamma]$ to the cycle determined by γ descends to an isomorphism

$$\begin{tikzcd} H_1(M) & \pi_1(M,q) \arrow[l, "\varphi"'] \arrow[d] \\ & \pi_1(M,q)_{ab} \arrow[ul, "\sim"] \end{tikzcd}$$

Here $\pi(M,q)_{ab}$ is the abelianization of $\pi(M,q)$. Define the map as in Theorem 17.7 $\Phi : H^1_{dR}(M) \to \mathrm{Hom}(\pi_1(M,q),\mathbb{R})$ by

$$\Phi[\omega][\gamma] = \int_{\tilde{\gamma}} \omega$$

where $[\gamma] \in \pi_1(M,q)$ and $\tilde{\gamma} \in [\gamma]$ is a piecewise smooth representative. Note that $\mathrm{Hom}(\pi_1(M,q)_{ab},\mathbb{R}) \cong \mathrm{Hom}(\pi_1(M,q),\mathbb{R})$ since the functional $\int_\bullet \omega$ kills elements of the form $\gamma \cdot \eta \cdot \gamma^{-1} \cdot \eta^{-1}$, where multiplication is concatenation and the inverse is the reversed path. Hence we an consider Φ as a homomorphism $H^1_{dR}(M) \to H^1(M)$, in which case it is nothing more than the deRham isomorphism. So we're done by deRham's theorem.

Problem 18-7. Let M be an oriented smooth n-manifold. Define $PD : \Omega^p(M) \to \Omega^{n-p}_c(M)^*$ by

$$PD(\omega)(\eta) = \int_M \omega \wedge \eta.$$

(a) Assuming M is closed, PD descends to a map on cohomology since

$$\begin{aligned} \int_M \mathrm{d}\xi \wedge \eta &= \int_M \mathrm{d}(\xi \wedge \eta) - \int_M (-1)^{\deg \xi} \xi \wedge \mathrm{d}\eta \\ &= (-1)^{\deg \xi + 1} \int_M \xi \wedge \mathrm{d}\eta \end{aligned}$$

using Stokes theorem. If η is closed then the integral vanishes.

(b)

Problem 18-8. (a) Since M is compact the compactly supported cohomology groups are isomorphic to the ordinary cohomology groups. First suppose M is orientable. By Poincaré duality, $H^p_{dR}(M) \cong H^{n-p}_{dR}(M)^* \cong H^p_{dR}(M)^{**}$. This is only possible if the de Rham groups are finite-dimensional, since in general if V is an infinite-dimensional vector space, then V is not isomorphic to V^{**}. Thus all the deRham groups are finite-dimesional in this case.

If M is not orientable, let $\pi : \widetilde{M} \to M$ be the orientation covering. Note that $\widetilde{M}$ is oriented and compact, since it is a two-sheeted covering of M. By Lemma 17.33, the induced map on cohomology $\pi^* : H^p_{dR}(M) \to H^p_{dR}(\widetilde{M})$ is injective. It follows from the first part that the de Rham groups are finite dimensional in this case as well.

(b) From the previous part we have

$$\dim H^p_{dR}(M) = \dim H^{n-p}_{dR}(M)^* = \dim H^{n-p}_{dR}(M).$$

Problem 18-9. When the de Rham groups are all finite dimensional, then the Euler characteristic

$$\chi(M) = \sum_{p=0}^{n} (-1)^p \dim H^p_{dR}(M)$$

is a homotopy invariant, since the de Rham groups are. If M is compact, orientable, and $2n+1$ dimensional then

$$\chi(M) = \sum_{p=0}^{(n+1)/2} (-1)^p \dim H^p_{dR}(M) + (-1)^{2n+1-p} H^{n-p}_{dR}(M) = 0.$$

Problem 19-1. Let $\mathcal{I} \subset \Omega^\bullet(M)$ be the set of all smooth differential forms which annihilate the distribution D. $\mathcal{I}$ is clearly an ideal in the sense of graded algebras. We claim that D is involutive if and only if $\mathcal{I}$ is a differential ideal, i.e. $\omega \in \mathcal{I}$ implies $\mathrm{d}\omega \in \mathcal{I}$. By Theorem 19.7, D is involutive if and only if for every η a 1 form annihilating D on an open set U, $\mathrm{d}\eta$ also annihilates D on U. From this it is clear that $\mathcal{I}$ being a differential ideal implies D is involutive, for we can extend any such η to 0 outside of U by multiplying by a smooth bump function ψ, since

$$\mathrm{d}(\psi\eta) = \mathrm{d}\psi \wedge \eta + \psi \mathrm{d}\eta$$

still annihilates D. The converse is trivial.

Problem 19-2. Let $\omega^1 \dots \omega^{n-k}$ be local defining forms for D. If D is involutive then $\mathrm{d}\omega^i = \sum_{i=1}^{n-k} \omega^i \wedge \alpha^i_j$ by Proposition 19.8, so

$$\mathrm{d}\omega^i \wedge \omega^1 \wedge \cdots \wedge \omega^{n-k} = 0, \qquad i = 1, \dots, n-k. \tag{5}$$

Conversely, if (5) holds then $\mathrm{d}\omega^i$ must annihilate D so D is involutive by Proposition 19.8.

Problem 20-4. We calculated in problem 8-28 that $\mathrm{d}(\det) = \mathrm{trace}$, so that the trace is the Lie algebra homomorphism induced by $\det : GL(n, \mathbb{R}) \to \mathbb{R}^\times$. By Proposition 20.8 the following diagram commutes:

$$\begin{tikzcd} \mathfrak{gl}(n, \mathbb{R}) \arrow[r, "\mathrm{tr}"] \arrow[d, "\exp"'] & \mathbb{R} \arrow[d, "\exp"] \\ GL(n, \mathbb{R}) \arrow[r, "\det"] & \mathbb{R}^\times \end{tikzcd}$$

Hence we obtain $\det e^A = e^{\mathrm{tr}(A)}$.

Problem 20-5. $A = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$ generates the subgroup $\exp tA = \begin{pmatrix} e^{ta} & 0 & 0 \\ 0 & e^{tb} & 0 \\ 0 & 0 & e^{tc} \end{pmatrix}$. $B = \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix}$ is step 3 nilpotent ($B^3 = 0$) so we calculate

$$\exp tB = I + tB + \frac{1}{2} t^2 B^2 = \begin{pmatrix} 1 & at & bt + \frac{1}{2}act^2 \\ 0 & 1 & ct \\ 0 & 0 & 1 \end{pmatrix}.$$

For $C = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ we note that $C^3 = -C$. Hence

$$\begin{aligned} \exp tC &= \sum_{j=0}^{\infty} \frac{t^{2j}}{(2j)!} C^{2j} + \sum_{j=0}^{\infty} \frac{t^{2j+1}}{(2j+1)!} C^{2j+1} \\ &= \cos t \cdot C^2 + \sin t \cdot C \\ &= \begin{pmatrix} -\cos t & \sin t & 0 \\ -\sin t & -\cos t & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Problem 20-6. (a) If $A \in SL(n, \mathbb{R})$ is of the form e^B for $B \in \mathfrak{gl}(n, \mathbb{R})$, then it has a square root $C = e^{B/2}$. $C \in SL(n, \mathbb{R})$ since $\det C = e^{\operatorname{tr}(B/2)} = e^0 = 1$.

(b) Set

$$A = \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & -2 \end{pmatrix}.$$

A is not in the image of $\exp : \mathfrak{sl}(2, \mathbb{R}) \to SL(2, \mathbb{R})$, otherwise it would have a square root in $SL(2, \mathbb{R})$. But it does not even have a real square root.

Problem 20-10. Let G be a Lie group and S a Lie subgroup. If S is closed then it is a properly embedded Lie subgroup (closed subgroup theorem). Otherwise, $\bar{S}$ is a properly embedded Lie subgroup and S is a dense subset of it.

Problem 21-18. Let G be a connected Lie group and let $H \subset G$ be a discrete normal subgroup. Then $H \subset Z(G)$, where $Z(G)$ is the center of G. To see this, for each $h \in H$ consider the map $C_h : G \to H$ given by $C_h(g) = hgh^{-1}g^{-1}$. C_h is clearly a diffeomorphism, and maps into H since $gHg^{-1} = H$. $C_h(G)$ is connected since G is, but $e \in C_h(G)$ so $C_h(G) = e$. This shows that $hg = gh$ for all $g \in G$ since H has the discrete topology. As h was arbitrary, the claim follows.

Problem 21-19. If G is a Lie group let $\pi : \widetilde{G} \to G$ be its universal covering group. We use the fact that $\pi_1(G) = \pi_1(G, e)$ is isomorphic to the deck transformations $\operatorname{Aut}_\pi(G)$. Recalling the construction of the universal covering group, we see that an element of $\operatorname{Aut}_\pi(G)$ is detemined by sending $\tilde{e}$, the identity of $\widetilde{G}$, to another element in the fiber $\pi^{-1}(e)$. Thus we can consider $\operatorname{Aut}_\pi(G)$ isomorphic to $H = \pi^{-1}(e)$. It is easy to see that H is a normal subgroup of $\widetilde{G}$ and that it has the discrete topology. By the result of Problem 21-18, H is central. In particular, it is abelian.

Problem 21-23. Let $A \in GL^+(n, \mathbb{R})$, the subgroup of invertible matrices with positive determinant. We recall the QR decomposition, which is obtained by using Gram-Schmidt

on the column vectors $a_1, \dots, a_n$ of A.

$$\begin{aligned}
b_1 &= a_1, & e_1 &= \frac{b_1}{|b_1|} \\
b_2 &= a_2 - (a_2 \cdot e_1)e_1, & e_2 &= \frac{b_2}{|b_2|} \\
&\vdots \\
b_n &= a_n - (a_n \cdot e_1)e_1 - \cdots - (a_n \cdot e_{n-1})e_{n-1} & e_n &= \frac{b_n}{|b_n|}.
\end{aligned}$$

In matrix form, $A = QR$ where $Q = (e_1, \dots, e_n)$ and $R = \begin{pmatrix} a_1 \cdot e_1 & a_2 \cdot e_1 & \dots & a_n \cdot e_1 \\ 0 & a_2 \cdot e_2 & \dots & a_n \cdot e_2 \\ 0 & 0 & \dots & a_n \cdot e_3 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \cdot e_n. \end{pmatrix}$.
This factorization is unique and $Q \in O(n)$ by construction. Note that

$$\det A = \det Q \prod_{i=1}^{n} a_i \cdot e_i > 0.$$

And since $a_i \cdot e_i = |b_i|$, it follows that $\det Q = 1$. This correspondence gives a map $f : GL^+(n, \mathbb{R}) \to SO(n) \times \mathbb{R}^{n(n+1)/2}$ sending A to $(Q, \mathbf{r})$ where $\mathbf{r} = (a_1 \cdot e_1, a_2 \cdot e_1, \dots, a_n \cdot e_n)$. This is obviously a diffeomorphism.

Problem 21-24. For $A \in GL(n, \mathbb{C})$ we follow the same procedure of problem 21-23, but instead we orthonormalize with respect the standard Hermitian inner product $z \cdot w = \sum z^i \bar{w}^i$. We get a factorization $A = QR$ where now $Q \in U(n)$ and we require that the diagonal entries of R are real and positive, by multiplying each e_i by a suitable unit complex number, if necessary. Therefore, we get a diffeomorphism $GL(n, \mathbb{C}) \approx U(n) \times \mathbb{C}^{n(n+1)/2-n} \times \mathbb{R}^n = U(n) \times \mathbb{R}^{n^2}$

Problem 21-25. Inspecting the construction in problem 21-23, if $A = QR \in SL(n, \mathbb{R})$ then $\det R = 1$. Therefore, the map f defined there restricts to $f : SL(n, \mathbb{R}) \to SO(n) \times \mathbb{R}^{n(n+1)/2-1}$, since the nth diagonal entry of R is determined after $n-1$ diagonal entries are chosen. By a similar consideration, $SL(n, \mathbb{C})$ is diffeomorphic to $SU(n) \times \mathbb{R}^{n^2-1}$. This shows that $SL(n, \mathbb{C})$ and $SL(n, \mathbb{R})$ are connected since $SO(n)$ and $SU(n)$ are. In particular, $SL(2, \mathbb{C}) \approx \mathbb{S}^3 \times \mathbb{R}^3$ is simply connected.

Problem 21-26. Problem 21-25 shows $SL(2, \mathbb{R}) \approx SO(2) \times \mathbb{R}^2 \approx \mathbb{S}^1 \times \mathbb{R}^2$, which has fundamental group $\mathbb{Z}$. Let $\pi : \widetilde{SL(2, \mathbb{R})} \to SL(2, \mathbb{R})$ be the univeral covering map and let $\rho : G \to GL(n, \mathbb{R})$ be any representation. We claim that ρ is not faithful (injective). Let $\rho_* : \mathfrak{sl}(2, \mathbb{R}) \to \mathfrak{gl}(n, \mathbb{R})$ be the induced Lie algebra morphism. Define $\varphi : \mathfrak{sl}(2, \mathbb{C}) \to \mathfrak{gl}(n, \mathbb{C})$ by

$$\varphi(A + iB) = \rho_* A + i\rho_* B, \qquad A, B \in \mathfrak{sl}(2, \mathbb{C}).$$

(a) φ is a Lie algebra morphism since

$$\begin{aligned}\varphi([A+iB, C+iD]) &= \rho_*([A,C]-[B,D]) + i\rho_*([A,D]+[B,C]) \\ &= [\rho_* A, \rho_* C] + i[\rho_* B, \rho_* C] + i[\rho_* A, \rho_* D] + i^2[\rho_* B, \rho_* D] \\ &= [\varphi(A+iB), \varphi(C+iD)].\end{aligned}$$

(b) Since $SL(2,\mathbb{C})$ is simply connected, it follows that there is a Lie group morphism $\Phi : SL(2,\mathbb{C}) \to GL(n,\mathbb{C})$ such that $\Phi_* = \varphi$. Moreover, the following diagram commutes.

$$\begin{tikzcd}
\widetilde{SL(2,\mathbb{R})} \arrow[d, "\pi"'] \arrow[dr, "\rho"] & \\
SL(2,\mathbb{R}) \arrow[d, hook, "\iota"'] & GL(n,\mathbb{R}) \arrow[d, hook] \\
SL(2,\mathbb{C}) \arrow[r, "\Phi"'] & GL(n,\mathbb{C})
\end{tikzcd}$$

(c) Hence, $\ker \rho$ is not trivial, since the kernel of $\Phi \circ \iota \circ \pi$ is not trivial (π covers $SL(2,\mathbb{R})$ countably many times).

Problem .